# Bellabeat: Smart Device Usage Analysis

Marketing Analytics Team | Prepared by Muhammad Zakaria Abbasi| Google Data Analytics Capstone

# Business task

## What Bellabeat wants to know

- **Task:** Analyze smart device usage data to find trends in how people use fitness trackers, and use them to recommend a marketing strategy for the Bellabeat app.
- **Stakeholders:** Urška Sršen (Co-founder, CCO), Sando Mur (Co-founder), Bellabeat marketing analytics team

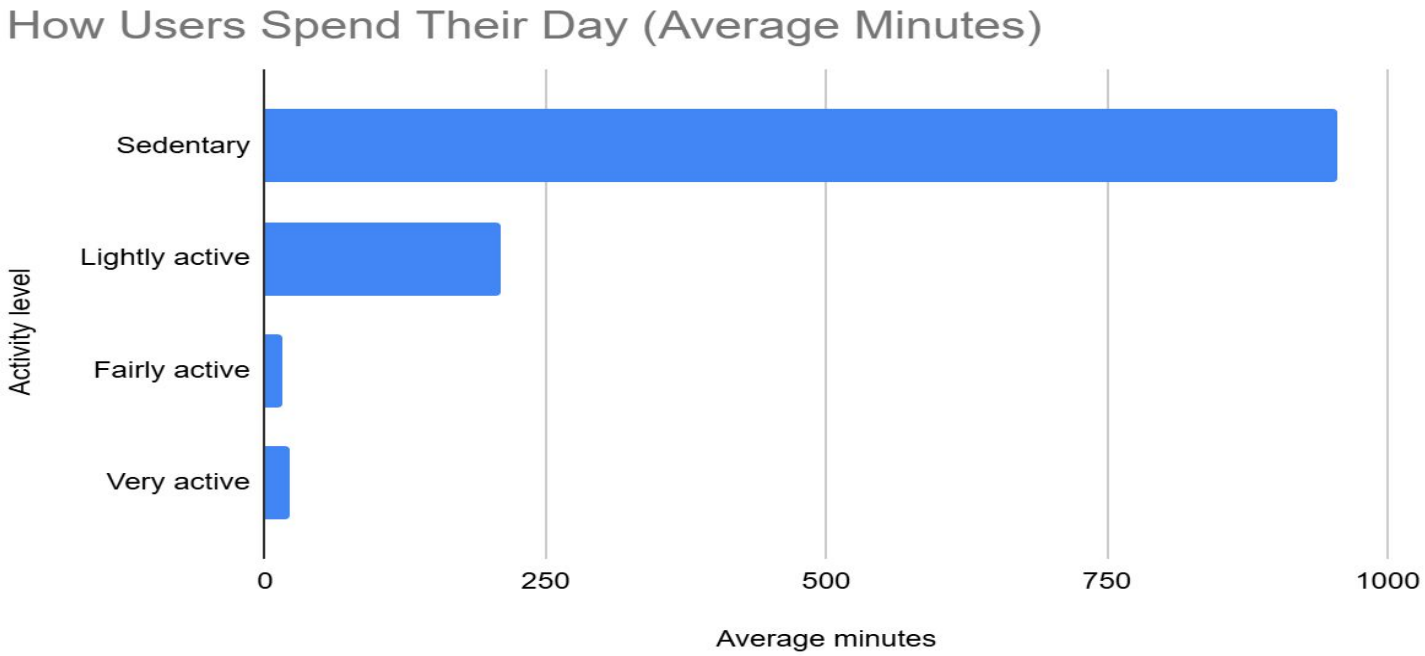
# Data and limits

## The data is useful but limited

- **Source:** Public FitBit Fitness Tracker dataset (Kaggle, CC0)
- **Sample:** 33 users in activity data, 24 in sleep data
- **Period:** 12 April to 12 May 2016 (31 days)
- **Limits:** Small sample, data from 2016, no gender or age information
- **Takeaway:** Treat the findings as directional and validate them with Bellabeat's own customer data.

# Finding 1

## Users are sedentary for about 16 hours a day

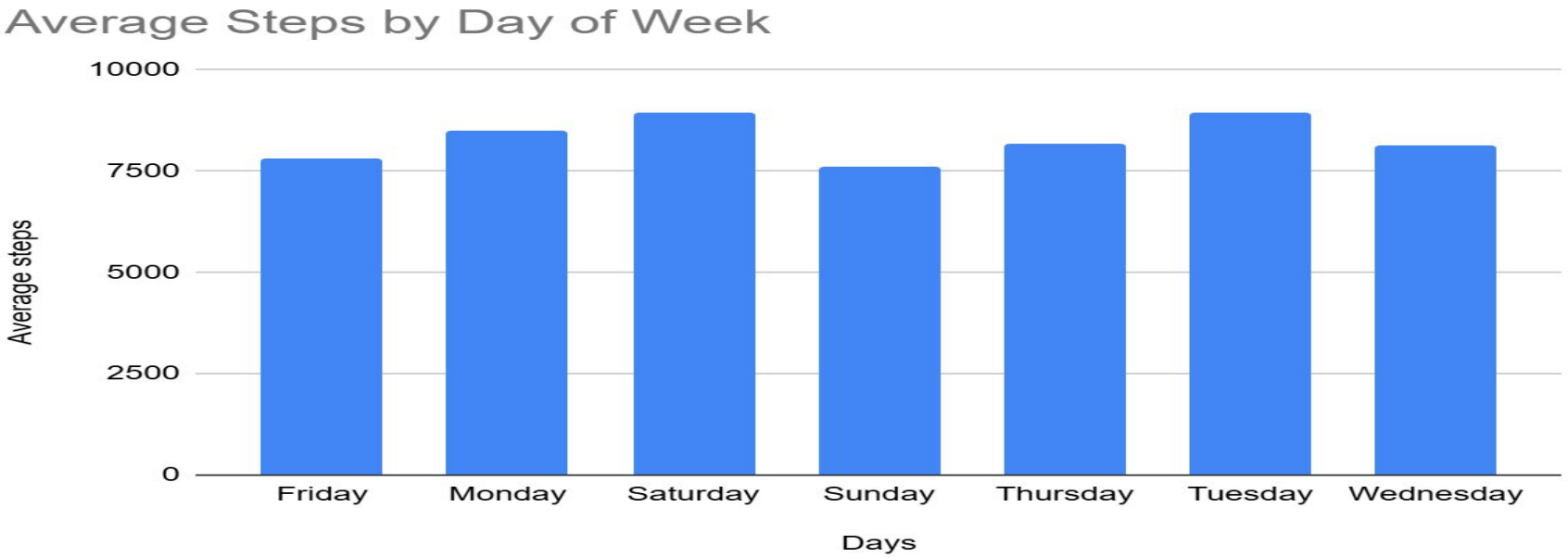


# Finding 1

- Sedentary: 956 min (about 15.9 hours)
- Lightly active: 210 min
- Fairly active: 15 min
- Very active: 23 min
- Average steps: 8,319, below the 10,000-step goal

# Finding 2

Tuesday and Saturday are the most active days, Sunday the least

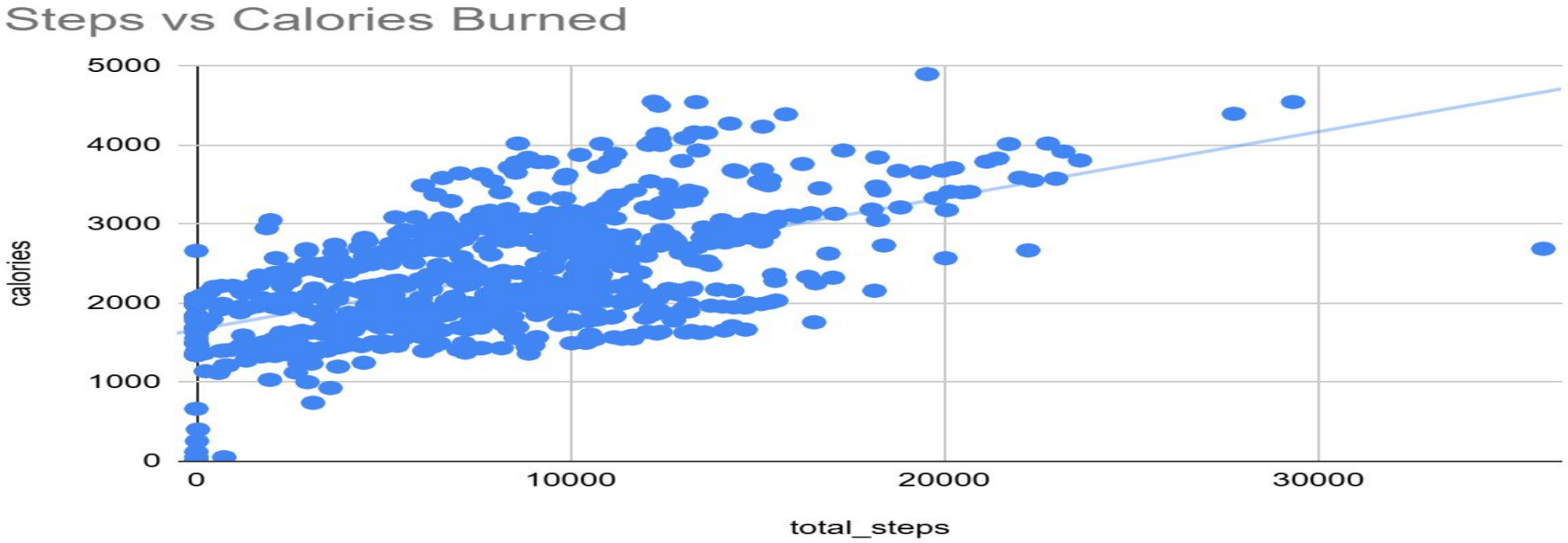


# Finding 2

- Tuesday: 8,949 steps, Saturday: 8,947 steps
- Sunday: 7,627 steps, Friday: 7,821 steps
- No day reaches 10,000 steps on average.

# Findings 3 and 4

More movement means more calories burned, and less sedentary time goes with better sleep



# Findings 3 and 4

- Steps and calories burned: moderate positive correlation (r = 0.56)
- Sedentary minutes and sleep: moderate-to-strong negative correlation (r = -0.60), among 24 users
- Users spend about 39 minutes a night in bed but not asleep (458 min in bed vs 419 min asleep).
- These are associations, not proof of cause.

# Recommendations for the Bellabeat app

1. **Send movement reminders** after long inactive periods, with stronger nudges on Fridays and Sundays.
2. **Build a daily step goal with streaks** and weekend challenges, since no weekday reaches 10,000 steps.
3. **Connect activity and sleep** with messages like “On days you move more, you tend to sleep better,” plus a bedtime reminder.

- **Next step:** Validate these findings with Bellabeat’s own larger, more recent customer data.